

24-502-0101: Mathematics for Computing

Core/Elective: **Core**

Semester: **1**

Credits: **4**

Course Description

This course introduces the study of mathematical structures that are fundamentally discrete in nature. The course is intended to cover the main aspects which are useful in studying, describing and modeling of objects and problems in the context of Linear Algebra, computer algorithms and programming languages.

Course Outcomes (CO)

After the completion of the course, the students will be able to:

CO	Couse Outcome Statement	Cognitive Level
CO1	Analyse the different methods for proving the correctness of the theorems and problems.	Analyse
CO2	Understand and apply the basic concepts of Linear Algebra.	Apply
CO3	Understand and apply the basic aspects of Descriptive statistics.	Apply
CO4	Understand and apply the fundamentals of probability theory.	Apply

Mapping with Program Outcomes

	PO1	PO2	PO3	PO4	PSO1	PSO2	PSO3	PSO4
CO1	3	-	-	-	3	3	-	2
CO2	3	3	-	-	2	2	-	2
CO3	3	3	-	-	2	2	-	2
CO4	3	3	-	-	3	2	-	2

Course Content

1. Introduction – proofs – propositions – predicates and quantifiers – truth tables – first order logic – satisfiability – pattern of proof – proofs by cases – proof of an implication – proof by contradiction – proving iff – sets – proving set equations – Russell’s paradox – well-ordering principle – induction – invariants – strong induction – structural induction
2. Vectors-Coordinate system-vector addition-vector multiplication-Linear combinations, span, and basis vectors-Matrix multiplication as composition-Three-dimensional linear transformations-The determinant-Inverse matrices, column space and null space- Nonsquare matrices as transformations between dimensions-Dot products and duality-Cross products-Cross products in the light of linear transformations-Cramer's rule-Change of basis-Eigenvectors and eigenvalues-vector spaces
3. Descriptive statistics: histogram, sample mean and variance, order statistics, sample covariance, sample covariance matrix – Frequentist statistics: sampling, mean square error, consistency, confidence intervals,

parametric and non-parametric model estimation

4. Probability theory: probability spaces, conditional probability, independence – Random variables: discrete and continuous random variables, functions of random variables, generating random variables – Multivariate random variables: joint distributions, independence, generating multivariate random variables, rejection sampling – Expectation: Mean, variance and covariance, conditional expectation
5. Random process: definition, mean and autocovariance functions, iid sequences, Gaussian and Poisson process, random walk – Convergence of random process: types of convergence, law of large numbers, Central limit theorem, monte carlo simulation – Markov chains: recurrence, periodicity, convergence, markov-chain monte carlo- Gibbs sampling, EM algorithm, variational inference

References

1. Bronson, R., Costa, G.B., Saccoman, J.T. and Gross, D., Linear algebra: algorithms, applications, and techniques. 4e, 2023.
2. Eric Lehman, F Thomson Leighton, Albert R Meyer, Mathematics for Computer Science, 1e, MIT, 2010.
3. Susanna S. Epp, Discrete Mathematics with Applications, 4e, Brooks Cole, 2010.
4. Gary Chartrand, Ping Zhang, A First Course in Graph Theory, 1e, Dover Publications, 2012. in
5. John Tsitsiklis. 6.041SC Probabilistic Systems Analysis and Applied Probability. Fall 2013. Massachusetts Institute of Technology: MIT OpenCourseWare. <https://ocw.mit.edu>
6. Albert Meyer. 6.844 Computability Theory of and with Scheme. Spring 2003. Massachusetts Institute of Technology: MIT OpenCourseWare, <https://ocw.mit.edu>.
7. Michael Mitzenmacher and Eli Upfal; Probability and Computing, 2ed, Cambridge University Press, 2017

Online Resources: Course notes of Carlos Fernandez-Granda, DS-GA 1002: Probability and Statistics for Data Science https://cims.nyu.edu/~cfgranda/pages/DSGA1002_fall17/index.html